

Assignment 5: Network Fundamentals and Building Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?

Answer: c) Forwarding data packets between networks

2. What is the purpose of DHCP (Dynamic Host Configuration Protocol)?

Answer: d) Dynamically assigning IP addresses to devices

3. Which network device operates at Layer 2 of the OSI model and forwards packets using MAC addresses?

Answer: b) Switch

4. Which network topology connects all devices linearly to a single backbone cable?

Answer: b) Bus

Section 2: True or False

5. True or False: A VLAN allows administrators to logically segment a single physical network into multiple virtual networks.

Answer: True

6. True or False: TCP is a connectionless protocol that provides reliable, ordered, and error-checked delivery.

Answer: False

7. True or False: A firewall monitors and controls incoming and outgoing network traffic based on security rules.

Answer: True

Section 3: Short Answer

8. Describe the steps involved in setting up a wireless network for a SOHO environment.

Answer:

Steps to set up a SOHO wireless network:

1. Choose a suitable wireless router

Select a router with dual-band support, WPA3 security, and required speed.

2. Connect the router to the modem

Use an Ethernet cable from the modem's LAN port to the router's WAN/Internet port.

3. Power on the router and modem

Wait for the devices to boot and LEDs to stabilize.

4. Access the router configuration page

Open a browser → enter the router's IP (e.g., **192.168.0.1** or **192.168.1.1**) → log in using default credentials.

5. Configure wireless settings

- a. Set SSID (Wi-Fi name)
- b. Select WPA2/WPA3 security
- c. Create a strong Wi-Fi password
- d. Choose appropriate channel/bandwidth

6. Set up DHCP

Enable DHCP so devices automatically receive IP addresses.

7. Configure additional security settings

- a. Disable WPS
- b. Enable firewall
- c. Change default admin password
- d. Optionally set MAC filtering

8. Save settings and reboot the router

Section 4: Practical Application

9. Demonstrate how to configure a router for Internet access using DHCP.

Answer:

To configure a router to obtain an Internet IP address using DHCP:

1. Connect router to modem

Plug modem → router WAN/Internet port.

2. Open router setup page

In a browser, enter the router's IP (e.g., 192.168.1.1).

3. Login to the admin interface

4. Navigate to Internet/WAN settings

5. Select "Connection Type: DHCP"

This allows the router to automatically receive:

- a. IP address
- b. Subnet mask

- c. Default gateway
- d. DNS server
- 6. Save settings
- 7. Reboot router and modem
- 8. Verify connection

Check WAN status—it should show a valid IP assigned by the ISP.

This completes DHCP-based router Internet configuration.

Section 5: Essay

10. Discuss the importance of network documentation in the context of building and managing networks.

Answer:

Network documentation is a critical component of building, maintaining, and managing any network infrastructure. It provides a complete record of how the network is structured, how devices are configured, and how different components interact. Without proper documentation, network management becomes inefficient, prone to errors, and difficult to troubleshoot.

1. Ensures Efficient Troubleshooting

When network issues occur, documentation such as topology diagrams, device configurations, and IP addressing schemes help technicians quickly locate problems. Instead of guessing, administrators can easily identify faulty components, incorrect settings, or misconfigured devices.

2. Supports Network Scalability and Future Upgrades

As organizations grow, they need to expand their networks.

Documentation acts as a blueprint that guides new installations and

prevents conflicts such as overlapping IP ranges or incompatible configurations. It ensures the network evolves in an organized and structured way.

3. Improves Security Management

Security policies, firewall rules, VLAN configurations, and access controls must be clearly documented. This helps administrators maintain consistent security settings across the network and identify vulnerabilities. It also supports compliance with security standards and audits.

4. Enhances Team Collaboration

Network administration often involves multiple team members. Documentation ensures that everyone has the same understanding of how the network is designed and managed. Even if a new technician joins, they can quickly understand the network without relying on verbal explanations.

5. Reduces Downtime

During outages, time is crucial. With detailed configuration records and device inventories, administrators can restore services faster. Backup configuration files help in quick recovery when devices fail.

6. Helps in Asset Management

Device lists, serial numbers, warranty information, and physical locations are all part of documentation. This makes it easier to manage hardware, schedule replacements, and track upgrades.

7. Ensures Compliance and Audit Readiness

Industries like banking, healthcare, and government require strict network documentation. Maintaining proper documentation helps meet legal requirements and avoids penalties.